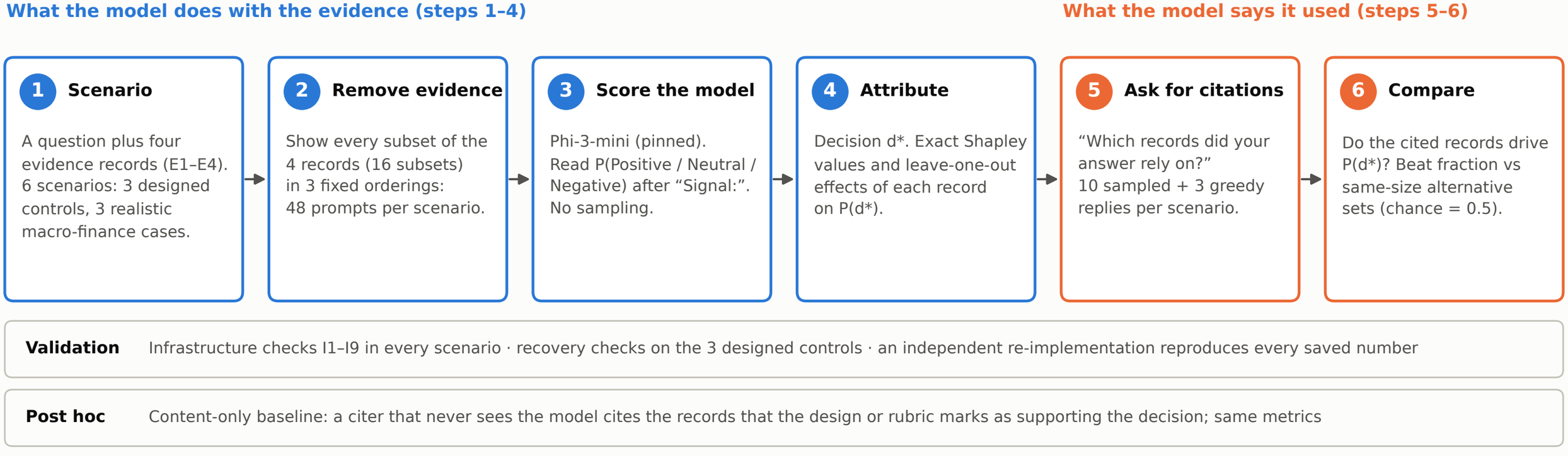
What the model does with the evidence (steps 1–4)
What the model says it used (steps 5–6)
1
Scenario
A question plus four evidence records (E1–E4). 6 scenarios: 3 designed controls, 3 realistic macro-finance cases.
2
Remove evidence
Show every subset of the 4 records (16 subsets) in 3 fixed orderings: 48 prompts per scenario.
3
Score the model
Phi-3-mini (pinned). Read P(Positive / Neutral / Negative) after “Signal:”. No sampling.
4
Attribute
Decision d*. Exact Shapley values and leave-one-out effects of each record on P(d*).
5
Ask for citations
“Which records did your answer rely on?” 10 sampled + 3 greedy replies per scenario.
6
Compare
Do the cited records drive P(d*)? Beat fraction vs same-size alternative sets (chance = 0.5).
Validation
Infrastructure checks I1–I9 in every scenario · recovery checks on the 3 designed controls · an independent re-implementation reproduces every saved number
Post hoc
Content-only baseline: a citer that never sees the model cites the records that the design or rubric marks as supporting the decision; same metrics